DAXHIEL PERUGORRIA RUCIEL

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EDUCATION

University of Texas at Austin – Austin, TX 2025 – Expected May 2029
Bachelor of Science in Computer Science

- **Relevant Coursework:** Data Structures and Algorithms, Computer Systems, Computer Organization and Architecture, Discrete Mathematics, Linear Algebra, Probability and Statistics

TECHNICAL SKILLS

- **Languages:** Python, Java, C/C++, Rust, JavaScript, SQL
- **Software Development:** FastAPI, WebSocket, asyncio, pytest, Express, Next.js, React, Tailwind CSS, Linux, Vercel, Supabase
- **Data & Machine Learning:** PyTorch, scikit-learn, LightGBM, XGBoost, Tableau, Data Visualization

EXPERIENCE

Code2College – Remote Jun 2026 – Jul 2026
Course Instructor

- Instructed 24+ students over an 18-session curriculum, with 95% average attendance and 97% average quiz score.
- Taught Python programming, object-oriented design, error handling, and basic data management.

Getfertilizerwiser – Austin, TX Feb 2026 – May 2026
Freelance Web Developer

- Built a full-stack Next.js/React web application for a nonprofit client, with a token-driven Tailwind design system and reusable component library keeping 7 client-facing pages visually consistent without a CMS.

Austin Spanish First AV Jul 2026 – Present
Subdirector

- Direct a 10+ member technical team coordinating audio/visual delivery for 400+ weekly attendees.
- Redesigned the audio and live-stream setup, growing online viewership by 50%.

Project Advance Austin – Austin, TX Aug 2026 – Present
Project Management & Data Analytics

- Migrating a nonprofit's fragmented Access/Excel program records into a consolidated database, deduplicating and normalizing records to support repeatable yearly updates (5-student team).
- Building a Tableau geospatial map of 2025 program data (children by county and state, mothers by prison facility, mother–child distance) to support donor fundraising.
- Own the build of a React/TypeScript site with a Node.js/Express and PostgreSQL backend that embeds the map and links from the client's site (repo, domain, SEO); due March 2027.

PROJECTS

qumixer-py | *Python Library for Allen & Heath Qu-series Mixers* Sep 2026

- Implemented the vendor MIDI-over-TCP protocol (corrected against real hardware) as sync and async APIs (QuMixer/AsyncQuMixer) with a tiered parameter registry.
- Built a fake console emulator with golden-vector fixtures for hardware-free testing of the encode/decode path, plus probe, monitor, and record/replay CLI tools.
- Debugged protocol mismatches with AI agents: a Claude agent ran tests on the physical mixer and reported gaps against the vendor reference for a second agent to fix.

Datahack 2026 | *Skip-Prediction Machine Learning Model* Mar 2026

- Team of 4 in UT Austin's MLDS hackathon; 1st of 5 finalists on the organizers' performance score; Transformer + LightGBM ensemble reached 86.50% validation accuracy vs. an 83.77% baseline.
- Co-designed the data pipeline and session tokenization; ran XGBoost and LSTM baselines.

VibeDungeon | *AI Agent-Driven Game* May 2026 – Jun 2026

- Solo-designed a Python dungeon crawler with an LLM agent layer isolated from core game logic, validating tool-call arguments against numeric ranges and enum allowlists; mostly written with AI tools.
- Replaced a generic agent with repetitive output by 6 specialized agents on a shared BaseAgent (tool loop, token limits, retries, fallbacks), provider-agnostic via the OpenAI SDK.